WW-T-700/2E
29 December 1983
SUPERSEDING
WW-T-700/2D
March 13, 1967

FEDERAL SPECIFICATION SHEET

TUBE, ALUMINUM, ALLOY, DRAWN, SEAMLESS, 3003

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring seamless tube drawn from aluminum alloy 3003 described herein shall consist of this document and the latest issue of WW-T-700/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for seamless tube drawn from aluminum alloy 3003.

1.2 Classification.

1.2.1 Tempers. The drawn seamless tube shall be of the following tempers: O, H12, H14, H16, H18, and F, as specified (see 6.2 and 6.3). The definitions of these tempers shall be as specified in WW-T-700/GEN.

1.2.2 Types. The tube shall be of the following types:

Type		Appearance
I	-	Round
II	-	Rectangular and square
III	-	Streamline
IV	-	Oval
V	-	Odd shapes

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specification

WW-T-700/GEN - Tube, Aluminum and Aluminum Alloy, Drawn, Seamless, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:26Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commerical Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Silicon	-	0.6
Iron	-	0.7
Copper	0.05	0.20
Manganese	1.0	1.5
Zinc	-	0.10
Other elements, each	-	0.05
Other elements, total 2/	-	0.15
Aluminum	Remainder	

1/ Except for "Aluminum" and "Others", analysis normally is made for elements for which specific limits are shown

2/ The sum of those "Others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:23Z
Check the source to verify that this is the current version before use.

3.2 Mechanical properties.

3.2.1 Tensile strength, yield strength and elongation properties. The tensile strength, yield strength and elongation mechanical properties parallel to the direction of drawing shall conform to the requirements specified in table II.

TABLE II. Tensile strength, yield strength and elongation properties

Temper	Wall thickness, inch	Tensile strength, minimum, ksi	Yield strength, minimum, ksi	Percent elongation in 2 inches or 4D 1/, minimum, kind of specimen	
				Full section	Cut-out
O	0.010-0.244 incl.	14.0	5.0	-	-
H12	0.010-0.599 incl.	17.0	12.0	-	-
H14	0.010-0.024 incl.	20.0	17.0	3	-
	0.260-0.500 incl.	20.0	17.0	-	-
H16	0.010-0.024 incl.	24.0	21.0	-	-
	0.260-0.500 incl.	24.0	21.0	-	-
H18	0.010-0.024 incl.	27.0	24.0	2	-
	0.260-0.500 incl.	27.0	24.0	-	-

1/ Round tube 2 inches or less in outside diameter and square tube 1-1/2 inches or less on a side shall be tested in full section unless the limitations of the testing machine preclude the use of such a specimen. For round tube over 2 inches in diameter, for square tube over 1-1/2 inches on a side, for all sizes of tube other than round or square, or in those cases when a full section specimen cannot be used, a cut-out specimen shall be used. D represents diameter of cut-out specimen.

2/ Maximum tensile strength is 19.0 ksi

3.2.2 Flattening. When specified (see 6.2), round tube (type 1) in O, H12, H14 and H16 tempers shall withstand, without cracking, the flattening test or the alternative bend test specified in WW-T-700/GEN. The values for flattening factor "F" are specified in table III.

TABLE III. Flattening factor

Temper	Wall thickness, inch	F
O	0.025 to 0.500, incl.	2
H12	0.025 to 0.259, incl.	3
H14	Up to 0.259, incl.	6
H16	0.025 to 0.259, incl.	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:23Z
Check the source to verify that this is the current version before use.

3.2.2.1 Alternative bending factor "N". The values for the alternative bending factor "N" are specified in table IV.

TABLE IV. Bending factor

Temper	N
H12	1
H14	4
H16	6

3.2.3 Flaring. When specified (see 6.2), round tube (type I) in the H14 temper shall be capable of being flared as specified in WW-T-700/GEN.

4. QUALITY ASSURANCE PROVISIONS (see WW-T-700/GEN)

5. PREPARATION FOR DELIVERY (see WW-T-700/GEN)

6. NOTES

6.1 Intended use. This tube is intended for use when very good formability, good weldability, or both are required, and when more strength is required than is provided by commercially pure aluminum tube. This tubing is not applicable for use in transferring hydraulic/pneumatic mediums.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Temper of material required (see 1.2.1)
(c) Type of tube required (see 1.2.2)
(d) Whether round tube (type 1) in O, H12, H14 and H16 tempers should withstand flattening (see 3.2.2)
(e) Whether round tube (type 1) in H14 temper should withstand flaring (see 3.2.3)
(f) Dimensions required
(g) Requirements for sizes not specifically covered (see WW-T-700/GEN)
(h) Selection of applicable levels of preservation and packing (see WW-T-700/GEN)

6.3 Availability of the types. Types II through V are only available in the O and F tempers. Type I is available in all tempers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:23Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST:

Custodians

Army-MR
Navy-AS
Air Force-20

Review Activities

Army-AR, EA
Navy-OS, SH
Air Force-99
DLA-CS

User Activities

Army-MI
Navy-MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
COMMERCE-NBS
HEW-FEC
NASA-KSC, MSF
USDA-AFS

PREPARING ACTIVITY:

NAVY-AS

DoD Project 4710-0708

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:23Z
Check the source to verify that this is the current version before use.